

ZATCA Fatoora Phase 2 E-Invoicing Compliance Architecture

SECTION 04-compliance-ksa TYPE policy STATUS active OWNER Compliance Lead / CTO

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This document establishes the technical, operational, and regulatory architecture for **ZATCA (Zakat, Tax and Customs Authority) Fatoora Phase 2 Integration** across BDA Company International's platforms (Fly GACA and Captain Adel).

1. Corporate & Tax Entity Identity

All invoices generated by Fly GACA B2B billing and consumer subscriptions strictly inherit from corporate facts in `01-governance/company-facts.md` :

Parameter	Certified Value
Legal Name (Arabic)	شركة بدع الدولية
Legal Name (English)	BDA Company International
Commercial Registration (CR)	7030976893
VAT Registration Number	311415259500003
Taxpayer Identification (TIN)	3114152595
Registered National Address	2816 King Fahd Rd, Al Sahafah Dist., Riyadh 13321-6548, Saudi Arabia
Standard Tax Rate	15% Standard VAT

2. Invoicing Classification & Flows

```
graph TD
  A[Billing Trigger: B2B Cohort / Stripe / Moyasar] --> B{Invoice Category}
  B -->|B2B Flight Academy / Enterprise| C[Standard Tax Invoice - 0100000]
  B -->|B2C Pilot Subscriptions / Prep Packs| D[Simplified Tax Invoice - 0200000]
  C --> E[Generate UBL 2.1 XML]
  D --> E
  E --> F[Canonical SHA-256 Digest Hashing]
  F --> G[ECDSA Cryptographic Signing with CSID]
  G --> H[Tag-Length-Value TLV QR Code Encoding]
  H --> I[ZATCA Fatoora Clearance / Reporting API]
  I --> J[Archived with Cryptographic Chaining PIH]
```

2.1 Standard Tax Invoices (B2B - Code 388 / 0100000)

- **Use Case:** Flight Academy bulk seat licenses (e.g., OxfordSaudia, SNAA, Saudi Aramco Aviation).
- **Clearance Workflow:** XML is transmitted to ZATCA Fatoora Clearance API before delivery to the customer. Once approved, the signed XML with clearance status is delivered.
- **Buyer Identifiers:** Mandatory Buyer Name, Buyer National Address, and Buyer VAT Number (where registered).

2.2 Simplified Tax Invoices (B2C - Code 388 / 0200000)

- **Use Case:** Individual student pilots purchasing Pro annual passes or AIP Exam Prep bundles.
- **Reporting Workflow:** Generated immediately with Phase 2 TLV QR code and reported asynchronously to ZATCA within 24 hours.

3. Cryptographic & Data Structures

3.1 Tag-Length-Value (TLV) QR Code (Phase 2 Specification)

The QR code embedded on all generated PDF and HTML invoices encodes a 9-tag Base64 binary payload:

1. **Tag 1:** Seller Name (BDA Company International)
2. **Tag 2:** Seller VAT Number (311415259500003)
3. **Tag 3:** Invoice Timestamp (YYYY-MM-DDTHH:MM:SSZ)
4. **Tag 4:** Invoice Total with VAT (SAR X.XX)
5. **Tag 5:** Total VAT Amount (SAR X.XX)
6. **Tag 6:** SHA-256 Invoice XML Digest
7. **Tag 7:** ECDSA Digital Signature (using secp256k1 / NIST P-256)
8. **Tag 8:** Seller Public Key (from CSID)
9. **Tag 9:** ZATCA Certificate Authority Signature

3.2 Cryptographic Chaining (Previous Invoice Hash - PIH)

Every invoice header includes a `<cac:AdditionalDocumentReference>` element containing the SHA-256 hash of the immediate preceding invoice (PIH). The genesis invoice uses a certified fixed seed hash

(NWZLY2ViNjZmZmM4NmYzOGQ5NTI3ODZjNmQ2OTZjNzljMmRiYzIzOWRkNGU5MWI0NjAzZTQ4MmUwNzMzNGZhNw==), establishing an immutable audit ledger compliant with anti-tampering mandates.

4. Implementation Reference

The production implementation resides in:

- Engine: `ay2m/FlyGACA/server/src/zatca-core.ts`
 - Automated Verification: `ay2m/FlyGACA/server/tests/zatca-core.test.ts` (4 unit tests verifying TLV encoding, SHA-256 digest, UBL 2.1 generation, and Academy B2B billing).
 - Tax Invoice Template: `ay2m/Office/03-finance/tax-invoice-template.html`
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5. Record Retention & Archival

Per ZATCA Article 66, all XML invoices, cryptographic logs, and verification responses must be archived in encrypted, in-Kingdom storage (Google Cloud Platform `me-central2` Dammam region) for a statutory period of **not less than 6 years (72 months)**.